CSE 340

NoSnooze

Wake up. Solve it. Start your day.



Meet the Team

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How most people start their morning:

A	**Alarm goes off**	Short-form content triggers dopamine spikes and crashes
B	**Tap snooze / dismiss**	Brain fog and reduced cognitive claraity persists for first few hours of day
C	**Open social media, Go back to sleep**	Going back to sleep is opposite of what we want when setting an alarm

Goatley, K. (2025). "Why Your Morning Scroll is Wrecking Your Mental Health" stillwaterstherapy.org

The Snooze Button Makes it Worse

Sleep inertia, the state of reduced arousal and impaired cognitive performance immediately upon waking is measurably worsened by repeated snoozing

"Snooze alarm use increases sleep inertia, thereby increasing fatigue" — Ogawa et al., 2022
Ogawa et al. (2022). "Snooze alarm use in a global population of smartphone users" pmc.ncbi.nlm.nih.gov/articles/PMC9804954

Our Solution

Replace the on-tap dismiss with a short puzzle

[1] MATH

Forces some cognitive function in the moments of waking.

[2] TRIVIA

Disrupts sleep inertia before reaching for social media.

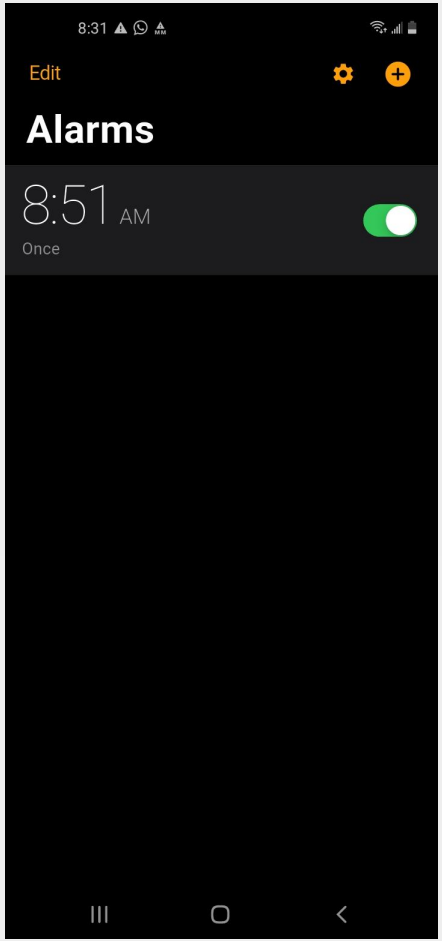
[3] PATTERN RECOGNITION

Beginning a healthy habit won't feel out of the way.

Application Demo

The Home Screen

- Add
- Edit
- Toggle
- Delete
- Settings -> Language
    - English and Spanish Support

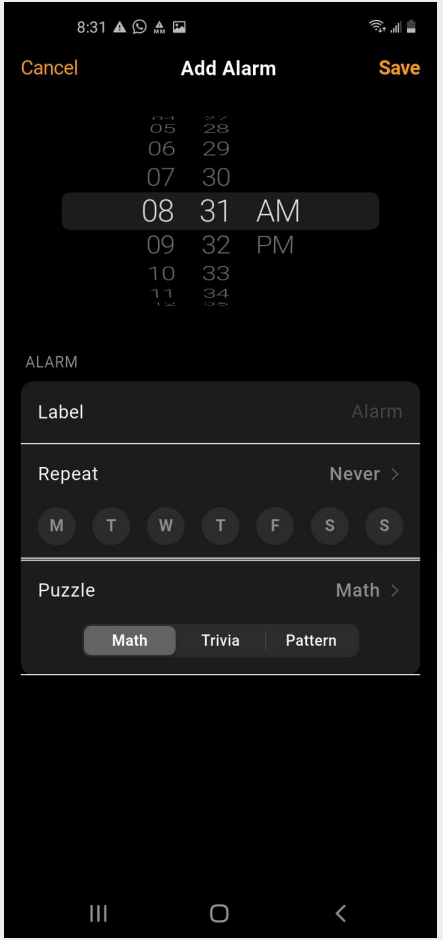




Application Demo

Add Alarm Screen

- Time Slider
- Custom Label
- Repeat Days
- Puzzle Choice
  - Math
  - Trivia (via API)
  - Pattern

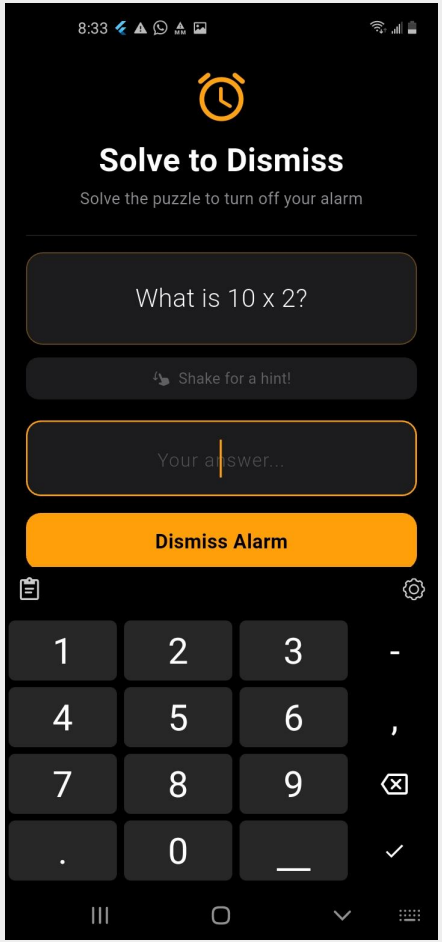




Application Demo

Alarm Dismiss Screen

- Math Puzzle Type
  - Arithmetic Prompt
  - Number answer
- Dismiss Alarm
- Shake / Swipe to reveal hint
- Alarms Persist





Usability Evaluation - SUS Questionnaire

Participant	SUS score
P1	85.0
P2	97.5
P3	72.5
P4	92.5
Mean	86.9

Usability Mean: 85.2, Learnability Mean: 93.8

Data Design

```
Alarm model (core data structure):
Alarm {
    Id: String (UUID)
    Label: String
    Hour: Int, minute: Int
    isEnabled: Bool
    repeatDays: Set<Weekday>
    puzzleType: PuzzleType (math|trivia|pattern)
}
```

Data Design

```
Puzzle model (transient - not persisted):
Puzzle {
    question: String
    answer: String
    hint: String?
    category: PuzzleCategory
}
```

Alarm is serialized to JSON -> stored in SharedPrefernces as a single string. Immutable model. All changes go through copyWith()

How Data Flows Through the App

User Action (add / edit / delete / toggle)

1 AlarmProvider

snapshot() undos stack
notifyListeners() UI rebuilds
persist() -> SharedPreferences
reschduleAll() -> alarm package

2 Alarm Fires (Alarm Package)

Alarm.ringing steam -> main.dart
Navigate to AlarmDismissScreen (UUID passed via payload)
AlarmProvider.getAlarmById(uuid)
PuzzleService.generate(puzzleType)
Puzzle displayed -> user answers -> alarm stopped

Techniques Used

- Sensors - accelerometer via sensors_plus, shake to reveal hint
- Web API - Open TrivaDB for trivia Puzzles, 5s timeout with local fallback
- Canvas - CustomePainter analog clock, trigonmetry for hand angles
- Guesture Detection - onPanUpdate + onPanEnd for swipe to reveal answer
- Undo/redo - Snapshot stacks in AlarmProvider,
- Data persistence - JSON serialisation to SharedPreferenes, restored before first frame
- Internationalisation - ARB files, flutter gen-10n EN + ES. locale is persisted across sessions

Challenges

- Alarm not navigating to puzzle screen, the Alarm.ringing stream emits a _CustomeHashSet<AlarmSettings> not a List and casting it as a list crashed silently, so the alarm would ring but dismiss screen did not appear. Fixed by casting to Iterable instead.

- Finding the right alarm model with screen opens: the alarm package only knows integer IDS and our model uses UUIDs. We first tried to reverse map via hashCode but that was unsuccessful so we then tried fixing it by storing theUUID directly in the AlarmSettings.payload at scheduling time then reading it from the stream and passing it straight to the dismss screen.

- Context access in initState: calling context.read<AlarmProvider>() from initState during a cold-start push cased provider lookup failures. We fixed this by deferring context access with await Future.microtask(() {}) so the first frame is guaranteed to be complete before reading from the tree.

What Changed

- Originally used flutter_local_notifactons for alarm but switch to alarm package (handles foreground service, audio, wake lock natively).

- Originally used integer alarm ID to dismiss screen but we decided to switch to passing UUID via payload field which allows for direct lookup and no fragile hashCode mapping.

- Originally had shake only for hint but added swipe down feature incase phone was unable to detect shaking.

- Originally english only but decided adding spanish localization might be beneficial.

Future Work

1. Difficulty settings: all puzzle as of right now are fixed difficulty but in the future we could add options to increase difficulty or maybe a dynamic difficulty depending on how user performs

2. Alarm history/streaks: app doesn't have memory of past alarms though we could implement something that tracks how long users take to dismiss alarms, how often user gets wrong/right answers, how much user uses hint feature etc., as well as adding a streak counter for getting answers right.

Thank You for Your Time!